The Chinese University of Hong Kong

Course Code: CSCI3170 Introduction to Database Systems

Mid Term Quiz 2025-2026 T2 (Time: 1 hour and 30 minutes)

1. Write your student ID on both the exam paper and the cover page of your answer book(s).
2. Submit both exam paper and answer book(s).
3. Please start each section in a new page.

STUDENT ID: _____

Section One [10 marks]

1. ***[2 marks]*** Briefly justify the needs for Derived Attributes, especially when they are practically ignored when translated to logical models.

2. For each of the following statements (1) answer Truth or False, and (2) provide a very short and simple explanation/example. Note 1: Explanations should be concise descriptions, and do not even need to be full sentences. Points will be lost for irrelevant info. Note 2: Each correct Truth or False statement is given one point, on that basis, two additional points are given for your explanation/example.
   (a) ***[4 marks]*** *"If an attribute in a relation is a primary key, it cannot be a foreign key at the same time."*
   (b) ***[4 marks]*** "Weak entities only participate in a 1 to many relationships with other entity(s) that own/identify it."

Section Two [50 marks]

For these following questions, consider the following schema. Reserves is a many to many relations between Sailors and Boats. Where Sid is the key of Sailors, Bid is the key of Boats. (Sid, Bid) is the key of Reserves; where sid, and bid are foreign keys pointing to the keys of Sailor and Boats respectively.

Sailors

Sid	Sname	Rating	Age
22	Dustin	7	45
29	Brutus	1	33
31	Lubber	8	55.5
32	Andy	8	25.5
58	Rusty	10	35
64	Horatio	7	35
71	Zorba	10	16
74	Horatio	9	40
85	Art	3	25.5
95	Bob	3	63.5

Boats

bid	bname	color
101	Interlake	blue
102	Interlake	red
103	Clipper	green
104	Marine	red

Reserves

sid	bid	day
22	101	1998-10-10
22	102	1998-10-10
22	103	1998-10-8
22	104	1998-10-7
31	102	1998-11-10
31	103	1998-11-6
31	104	1998-11-12
64	101	1998-9-5
64	102	1998-9-8
74	103	1998-9-8

3. Π_{Sname} ($\sigma_{bid=103}$ *(Reserves* ⋈ *Sailors))*

   (a) ***[4 marks]*** Express what the above expression does in plain English

   (b) ***[4 marks]*** Is ⋈ above a natural join or a theta join?

   (c) ***[4 marks]*** Given that ⋈ is an extended operator based on basic operators, rewrite the expression using basic operators only. In addition to retaining the original meaning, the output should be exactly the same.

4. Answer the following questions in relational algebra

   (a) ***[5 marks]*** Find Sname of sailors who reserved at least a blue boat, but never any green boats

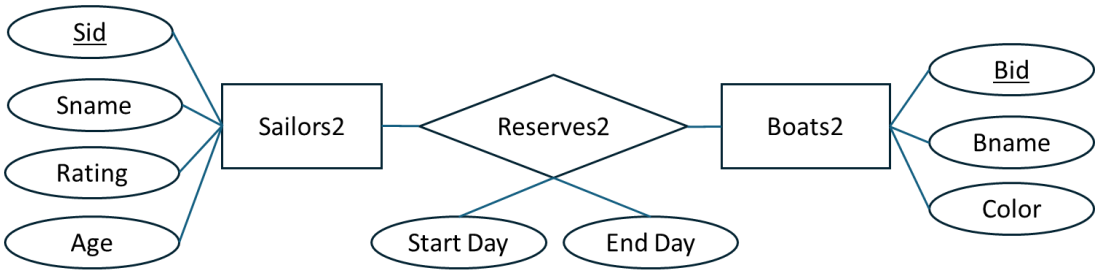
   (b) ***[5 marks]*** Find Sid and Sname of sailors who have never reserved any boats

   (c) ***[5 marks]*** Find Sids of sailors who has the greatest age

   (d) ***[6 marks]*** Find Sids, of sailors who have reserved a boat on the same day

   (e) ***[6 marks]*** Find Sid of sailors who have reserved exactly one boat in the Reserves

5. Consider a slightly updated ER diagram, and answer the question on the next page



(a) ***[3 marks]*** *What is main semantic difference between this and above schema?*

(b) ***[4 marks]*** *A person still cannot reserve the same boat twice or more in this conceptual model. Explain briefly why.*

(c) ***[4 marks]*** A student argues that we can theoretically, we can solve the issue in (b) by making the key of Reserves2 to be all its attributes. I.e. (Bid, Sid, StartDay, EndDay); another student argues that it should be only (Bid, Sid, StartDay). Which one is more appropriate, if so, why?

Section Three [20 marks]

Note: Please use the main drawing conventions in the lecture notes (for cardinality, use the course notation!) Do not use surrogate keys. If not specified, please note all assumptions that you make.

6. ***[10 marks]*** Draw an ER diagram to represent this scenario and clearly state the assumptions you make if any. Consider the following course-enrolment business application:

- A course is uniquely identified by its course ID. For each course, we also record these attributes: course name, lecturer name, and total capacity. We are interested in the total number of students who enrol in each course as well.
- Each course can have zero or more labs. A lab is uniquely identified by a lab ID. Each lab contains these attributes: time, location and weeks. Lab location is composed of a building name and a room number, and a lab can be held in multiple weeks.
- A student is uniquely identified by his/her sID. Each student also has the attributes: name, contact number, email and current degree.
- A lab tutor is uniquely identified by his/her ID. Each lab tutor also has the attributes: name, contact number, email and salary.
- A student can enrol in zero or more courses and can enrol in zero or more labs. Each lab must belong to exactly one course and must have at least one tutor on duty.
- A lab tutor can work for zero or more courses and zero or more labs.
- A course or a lab must have at least one student. A course can have multiple lab tutors but is not compulsory to have lab tutors.

7. ***[10 marks]*** Translate the ER diagram you drew above in question 7 into a relational diagram.

Section Four [20 marks]

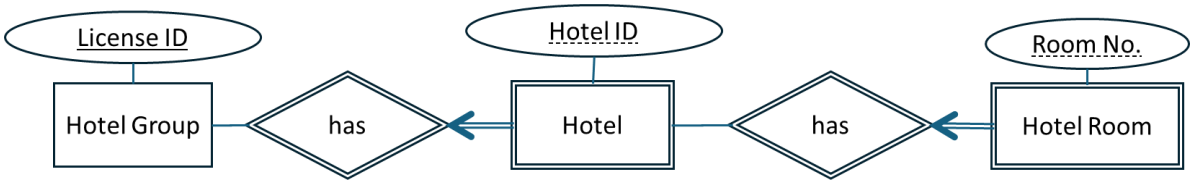
8. Note 1: Explanations should be concise descriptions, and do not even need to be full sentences. Points will be lost for irrelevant info. Note 2: Each correct Truth or False statement is given one point, on that basis, two additional points are given for your explanation/example.

(a) ***[4 marks]*** "A column that is a foreign key can contain duplicate values".

(b) ***[4 marks]*** "A column that is a foreign key cannot contain null values".

(c) ***[4 marks]*** A student claims that you can implement total-participation by via foreign-keys alone and nothing else. If achievable, explain with a small example; if not achievable, explain with a simple counter example.

9. In relation to the following Entity-Relation Diagram, answer relevant questions.



(a) ***[4 marks]*** Translate above ER diagram into a Relational diagram.

(b) ***[4 marks]*** *Based on the model, a student argues that a Hotel ID intuitively should naturally be unique and thus should instead be a strong entity. Comment on whether it is still the same scenario if Hotel were to be modelled instead with a strong entity.*

End of mid-term, good job